

# JEE MAIN REVISION MASTER: MATRICES AND DETERMINANTS

Subject: Mathematics | Target Scope: JEE Main Previous Years | Empirical Base: 65 Analyzed PYQs

## 1. High-Yield Concept Priority Matrix

| Priority Tier | Concept Name                                          | Frequency | Category               |
|---------------|-------------------------------------------------------|-----------|------------------------|
| Very High     | Determinants, Adjoint, Inverse & System of Equations  | 50 PYQs   | Determinants & Systems |
| Very High     | Matrix Algebra: Transpose, Symmetric & Skew-Symmetric | 16 PYQs   | Matrices               |

## 2. Master Formula Reference Card

| Concept / Topic                                       | Mathematical Formulation & Governing Equations                                                                                            |
|-------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------|
| Determinants, Adjoint, Inverse & System of Equations  | $ kA  = k^n  A $ , $ \text{adj } A  =  A ^{n-1}$ , $A^{-1} = \frac{\text{adj } A}{ A }$ , $ \text{adj } (\text{adj } A)  =  A ^{(n-1)^2}$ |
| Matrix Algebra: Transpose, Symmetric & Skew-Symmetric | $(AB)^T = B^T A^T$ , $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$ , $\text{Skew-symmetric: } a_{ii} = 0$                             |

## 3. Deep-Dive Concepts & Linked PYQ Evidence

### 3.1 Determinants, Adjoint, Inverse & System of Equations (Very High Yield — 50 PYQs)

**Core Principle:** Determinant properties:  $|kA| = k^n |A|$  for  $n \times n$  matrix. Adjoint properties:  $A \cdot (\text{adj } A) = |A| I$ ;  $|\text{adj } A| = |A|^{n-1}$ ;  $|\text{adj}(\text{adj } A)| = |A|^{(n-1)^2}$ ;  $\text{adj}(AB) = \text{adj}(B) \cdot \text{adj}(A)$ . Inverse:  $A^{-1} = (\text{adj } A / |A|)$  exists if and only if  $|A| \neq 0$  (non-singular). System  $AX = B$ : Unique solution if  $|A| \neq 0$ ; no solution (inconsistent) if  $|A|=0$  and  $(\text{adj } A)B \neq 0$ .

■ **Caution / JEE Pitfall:** For  $3 \times 3$  matrix ( $n=3$ ):  $|2A| = 2^3 |A| = 8|A|$ , NOT  $2|A|$ ! And  $|\text{adj } A| = |A|^{3-1} = |A|^2$ !

■ **Speed Trick:** Shortcut for  $2 \times 2$  matrix inverse: Swap diagonal elements, change signs of off-diagonal elements, divide by  $|A|$ .

#### Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "The area of the triangle with vertices  $(3,8), (-4,2)$  and  $(5,1)$  is  $(P/4)$ , then the value of  $P$  is..." [Ans: C]
- [2026 Shift] Q#1 (MCQ): "Consider the following statements: Statement I: If  $A$  is a non-singular matrix, then  $A^{-1}$  exists. Statement II: If  $A$  and  $B$  are symmetric ..." [Ans: N/A]
- [2026 Shift] Q#1 (MCQ): "Which of the following is correct?..." [Ans: C]

### 3.2 Matrix Algebra: Transpose, Symmetric & Skew-Symmetric (Very High Yield — 16 PYQs)

**Core Principle:** Matrix operations: Matrix multiplication is non-commutative ( $AB \neq BA$ ). Transpose properties:  $(AB)^T = B^T A^T$ . Symmetric matrix:  $A^T = A$  ( $a_{ij} = a_{ji}$ ). Skew-symmetric matrix:  $A^T = -A$  ( $a_{ij} = -a_{ji}$ ). All diagonal elements of a skew-symmetric matrix are strictly ZERO ( $a_{ii} = 0$ ). Any square matrix  $A$  can be uniquely decomposed as  $A = (A + A^T / 2) + (A - A^T / 2)$  (symmetric + skew-symmetric).

■ **Caution / JEE Pitfall:** The determinant of an odd-order skew-symmetric matrix is ALWAYS zero:  $|A| = 0!$

■ **Speed Trick:** If  $A$  is an idempotent matrix ( $A^2 = A$ ), then  $(I + A)^n = I + (2^n - 1)A$ .

**Representative Source PYQs:**

- **[2026 Shift] Q#2 (MCQ):** "Match List-I with List-II List-I List-II a) A matrix which is not a square matrix i) Symmetric matrix b) A square matrix  $A = A^T$  ii) Null ..." **[Ans: N/A]**
- **[2025 Shift] Q#1 (MCQ):** "If a matrix  $A = \begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix}$  satisfies  $A^6 = kA^4$ , then the value of  $k$  is..." **[Ans: N/A]**
- **[2025 Shift] Q#2 (MCQ):** "If  $A$  is a square matrix such that  $A^2 = A$ , then  $(I - A)^3$  is..." **[Ans: N/A]**

## 4. Benchmark Illustrative PYQs with Master Solutions

### Benchmark Problem 1: KCET 2026 (Q#1)

The area of the triangle with vertices  $(3,8)$ ,  $(-4,2)$  and  $(5,1)$  is  $(P / 4)$ , then the value of  $P$  is

- (A)  $(61 / 2)$
- (B)  $(2 / 61)$
- (C) 122 **[Correct]**
- (D)  $(1 / 122)$

**Correct Answer: C**

**Solution Analysis:** Using the area formula for a triangle with vertices  $(x_1, y_1)$ ,  $(x_2, y_2)$  and  $(x_3, y_3)$ ,  $\text{Area} = \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$ . Here the vertices are  $(3,8)$ ,  $(-4,2)$  and  $(5,1)$ . So,  $\text{Area} = \frac{1}{2} |3(2-1) + (-4)(1-8) + 5(8-2)| = \frac{1}{2} |3(1) + (-4)(-7) + 5(6)| = \frac{1}{2} |3 + 28 + 30| = \frac{1}{2} \times 61 = (61 / 2)$ . Given that the area is  $(P / 4)$ ,  $(P / 4) = (61 / 2)$ . Multiplying both sides by 4,  $P = 122$ . So, the correct option is 122 Option C.